

SCORE/30:

Math 171 (sec. 501)
Analytic Geometry & Calculus
Tuesday, February 5, 2008

NAME:

PRACTICE QUIZ 4 (with solutions)

Please show your work and write only in **pen**. Calculators are **not** allowed. Notes are **not** allowed. Please also note that there are exactly **5** problem.

1: Suppose $\frac{e^{2x}}{3e^{2x}-1} = y$ and $y \neq 1/3$. Please solve for x .

Seeing how e^{2x} is repeated, we can try setting $u = e^{2x}$ to break the problem into pieces. For then we have $\frac{u}{3u-1} = y$ and we can solve for u as a function of y , and **then** apply $x = \frac{1}{2} \log(u)$ to get x as a function of y .

To do the first, note that we have $u = (3u - 1)y$ and thus $(1 - 3y)u = -y$. So $u = \frac{y}{3y-1}$. Applying $x = \frac{1}{2} \log(u)$, we thus obtain

$$x = \frac{1}{2} \log \left(\frac{y}{3y-1} \right).$$

2:

(a) Please find a sequence of real numbers $(x_1, x_2, x_3, \dots)$ such that $\lim_{i \rightarrow +\infty} x_i = 0$ and, for all i , we have $\sin(1/x_i) = 1$.

(b) For your answer from (a), what i could give you $|x_i| < 10^{-6}$?

(c) For your answer from (a), what i could give you $|x_i| < 10^{-999}$?

(d) For your answer from (a), what i could give you $|x_i| < \delta$?

(a) Letting $y_i = 1/x_i$, one thing we certainly need to do is find y_i that gives us $\sin y_i = 1$. In particular, since x_i tends to 0, we could try to find y_i that, in addition, tend to $+\infty$. From basic trigonometry, we know that $\sin y = 1$ exactly when y is $\pi/2$ plus a multiple of 2π . In other words, we can take $y_i = \frac{\pi}{2} + 2i\pi$. (For then, indeed, $y_i \rightarrow +\infty$ as $i \rightarrow +\infty$ and $\sin y_i = 1$ for all i .)

So $\boxed{x_i := \frac{1}{\frac{\pi}{2} + 2i\pi}}$ works. **Note:** There are other correct answers, e.g., $x_i := \frac{1}{\frac{5\pi}{2} + 8(-i)^3\pi}$.

(b) For our answer from Part (a), x_i is always positive. So enforcing $|x_i| < 10^{-6}$ is the same as enforcing $x_i < 10^{-6}$. From our formula, we then need to enforce $\frac{1}{\frac{5\pi}{2} + 2i\pi} < 10^{-6}$, which is in turn equivalent to $10^6 < \frac{5\pi}{2} + 2i\pi$. A little more arithmetic then shows that the latter inequality is true if and only if $\boxed{i > \frac{10^6 - \frac{5\pi}{2}}{2\pi}}$.

(c) Performing almost exactly the same calculations as in Part (b), we obtain $\boxed{i > \frac{10^{999} - \frac{5\pi}{2}}{2\pi}}$.

(d) Performing almost exactly the same calculations as in Part (b), we obtain $\boxed{i > \frac{\frac{1}{\delta} - \frac{5\pi}{2}}{2\pi}}$.

The point of Parts (b)-(d) is simply to show that working with δ is no harder than working with a number.

3: Is it possible for a function f to have $\lim_{x \rightarrow 5^-} f(x)$ and $\lim_{x \rightarrow 5^+} f(x)$ well-defined, but $\lim_{x \rightarrow 5} f(x)$ not well-defined? Please prove your answer either way. Hint: If you claim there is such an f , exhibit one. If you claim there is none, prove that the existence of such an f leads to a contradiction.

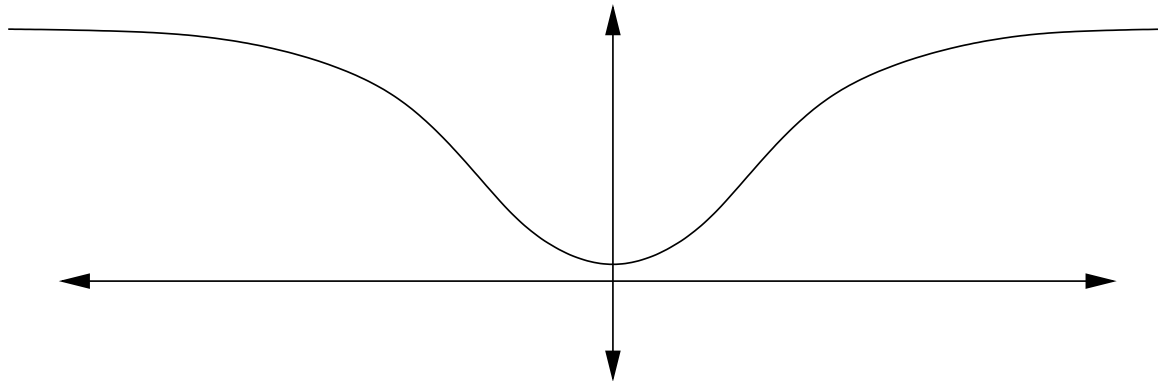
The answer is **YES**, and there are many correct examples. The key point is that one could have the left and right hand limits well-defined but **distinct**. For example, $f(x) := \frac{x-5}{|x-5|}$ has $\lim_{x \rightarrow 5^-} f(x) = 1$ but $\lim_{x \rightarrow 5^+} f(x) = -1$, so $\lim_{x \rightarrow 5} f(x)$ can not be well-defined.

4: Can you find a function f that is discontinuous at 1 but differentiable at 1? Please prove answer either way. Hint: If you claim there is such a function, exhibit one. If you claim there is none, prove that the existence of such a function leads to a contradiction.

The answer is **NO**, and the proof follows:

Suppose, to derive a contradiction, that we did have a function f with f discontinuous at 1 but differentiable at 1. Then $\lim_{h \rightarrow 0} \frac{f(1+h)-f(1)}{h} = L$ for some real number L , i.e., the limit defining the derivative must be well-defined. But then, since $\lim_{h \rightarrow 0} h = 0$, we must have $\lim_{h \rightarrow 0} \left(h \frac{f(1+h)-f(1)}{h} \right) = \lim_{h \rightarrow 0} (f(1+h) - f(1)) = 0$. In other words, $\lim_{h \rightarrow 0} f(1+h) = f(1)$, and f must therefore be continuous at 1.

5: A function f has a graph resembling the following curve:

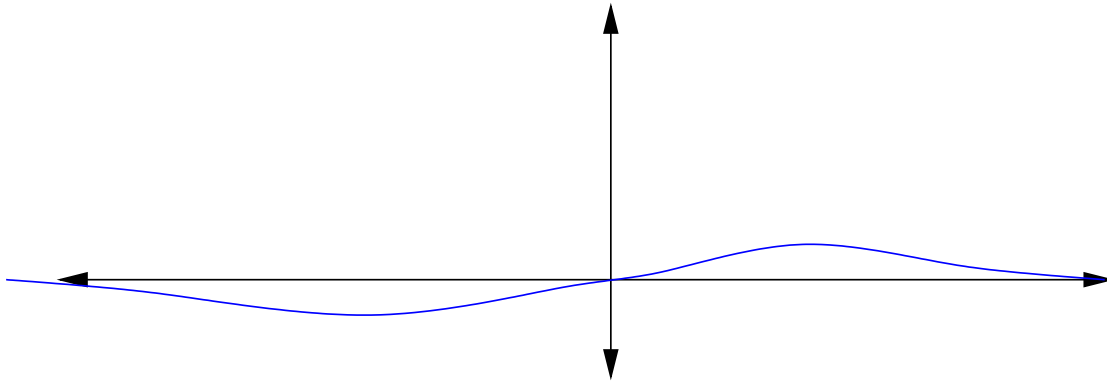


Please sketch the graph of the derivative of f . (The key data you need to depict correctly are the sign changes and asymptotes.)

The key facts to remember are:

1. On an interval of increase, a (differentiable) function has **positive** derivative.
2. On an interval of decrease, a (differentiable) function has **negative** derivative.
3. On an interval of constancy, a (differentiable) function has **zero** derivative.

Bearing this in mind, one correct answer would be:



The only acceptable variation among answers would be the depth of the valley and height of the peak in the figure above: No matter what is drawn, it must be clear that $f'(x)$ goes from 0 to negative to 0 (near $x=0$) to positive to 0, as x goes from $-\infty$ to ∞ .

6: Please find the derivative of $f(x) = (x^7 - \sin x)e^x$.

By the product rule, we have $f'(x) = \frac{d}{dx}(x^7 - \sin x) \cdot e^x + (x^7 - \sin x) \frac{d}{dx}(e^x) = (7x^6 - \cos x)e^x + (x^7 - \sin x)e^x = (x^7 + 7x^6 - \sin(x) - \cos(x))e^x$. So the answer is $\boxed{(x^7 + 7x^6 - \sin(x) - \cos(x))e^x}$.

7: Let S be the set of all cars, let A be the phrase
“ x is a Porsche”

and let B be the phrase

“ x is a red”

Which of the following are true? Please explain either way.

- (a) $\exists x \in S [A \wedge B]$
- (b) $\forall x \in S [A \implies B]$

(a) This sentence, in English, becomes “There is a car that is a red Porsche.” To the best of my knowledge, it is indeed true, for I have seen red Porsches (my father had one).

(b) This sentence, in English, becomes “Every car that is a Porsche is red.” To the best of my knowledge, the sentence is false, for I have seen non-red Porsches.

8: Is there a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that f is continuous at 0 but discontinuous everywhere else? Hint: If you claim there is such a function, exhibit one. If you claim there is none, prove that the existence of such a function leads to a contradiction.

The answer is YES. One example comes from our text: Take $f(x)$ to be 0 or x , according as x is rational or not. Then f is discontinuous at every $x \neq 0$ but f is continuous at 0.

Intuitively, the reason for this is that f is always oscillating, but this oscillation damps down to 0 at $x=0$. Rigorously, the proof would proceed as follows:

(Discontinuity Away from $x=0$) If x is a nonzero rational number, then one can always find a sequence of **irrational** numbers $(x_1, x_2, x_3, \dots)$ such that $\lim_{i \rightarrow +\infty} x_i = x$. (For instance, $x_i := x + \frac{\pi}{i}$.) Since the x_i are irrational, we obtain $f(x_i) = x_i > 0 = f(x)$ for all i . The equality $\lim_{h \rightarrow 0} f(x+h) = f(x) = 0$ is therefore impossible.

If, on the other hand, x is a nonzero irrational number, we can still build a similar sequence — rational instead — and proceed almost identically.

(Continuity at $x=0$) Here, we can simply use the definition of continuity: proving that f is continuous at 0 is the same as proving that $\lim_{x \rightarrow 0} f(x) = f(0)$. Now, since $|f(x)| \leq |x|$ for all x by the definition of f , we immediately obtain that $\lim_{x \rightarrow 0} |f(x)| \leq \lim_{x \rightarrow 0} |x|$. Since the latter limit is clearly 0, we thus obtain that $\lim_{x \rightarrow 0} |f(x)| = 0$. In particular, $\lim_{x \rightarrow 0} f(x) = 0$ and we're done.